

Amal Jyothi College of Engineering (Autonomous)

Amal Jyothi Scholars Community for Inspiring Innovation (ASCII)

R & D arm of IEDC

20th ASCII I2U 2026 Project Contest

1. TITLE OF THE PROJECT: Hybrid Vision Transformer-U-Net Framework for Automated Skin Cancer Segmentation, Classification, and Clinical Decision Support

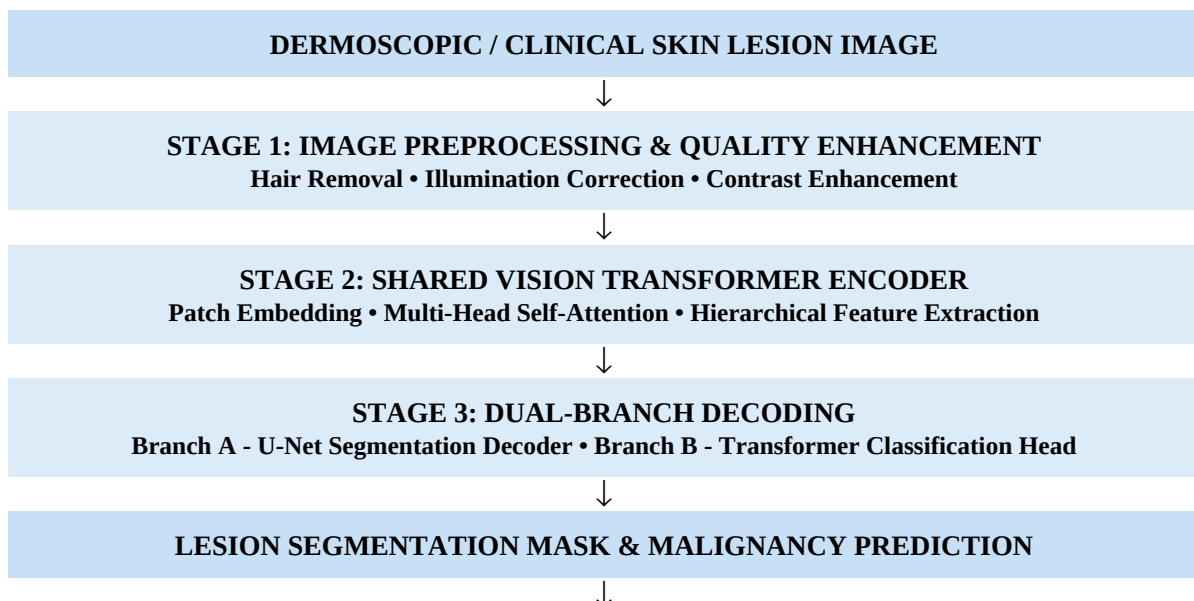
2. PROJECT SYNOPSIS (100 WORDS):

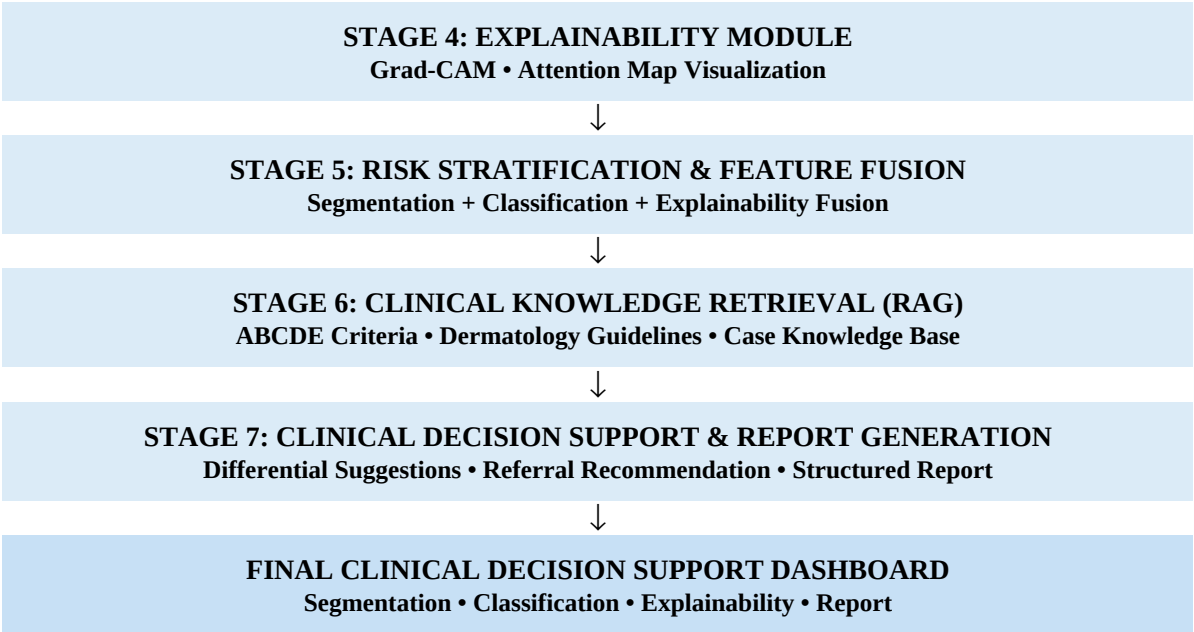
This project proposes a hybrid Vision Transformer-U-Net framework for automated skin lesion segmentation, classification, and clinical decision support within a single pipeline. The work builds on a recent hybrid Transformer-CNN framework that uses separate pathways for classification and segmentation and already incorporates Grad-CAM explainability. This project modifies that approach by introducing a shared Vision Transformer encoder with two task-specific branches - a U-Net-style decoder for lesion segmentation and a transformer classification head for malignancy prediction - enabling joint feature learning. Grad-CAM explainability and retrieval-augmented clinical knowledge integration further support interpretable, guideline-linked decision support for dermatology screening.

3. POTENTIAL APPLICATIONS:

The proposed system can support dermatology clinics and hospitals in faster, more consistent triage of suspicious skin lesions, and can be extended to teledermatology and mobile screening platforms for early detection in low-resource or rural settings where dermatologist access is limited. The explainable, guideline-linked decision support can assist general physicians and nurses in prioritizing referrals for suspected malignant lesions. The framework can also serve as a research and educational tool for studying interpretable multi-task learning in medical imaging, and as a foundation for future integration into hospital screening workflows and public-health skin cancer awareness programs.

4. GENERAL PLAN OF IMPLEMENTATION (Block diagram, flow chart etc.):





The framework integrates lesion segmentation and classification through a shared Vision Transformer encoder, avoiding the redundant feature extraction of separate-pathway baselines. Task-specific branches decode lesion boundaries and predict malignancy, while Grad-CAM-based explainability and retrieval-augmented clinical knowledge integration convert model outputs into interpretable, guideline-linked decision support for dermatology screening.

5. PROJECT TIMELINE WITH MILESTONES:

Phase	Period	Major Activities	Milestone
Phase 1	Month 1	Literature review, dataset acquisition (ISIC 2019/2020, HAM10000), preprocessing pipeline, Google Drive-Colab environment setup	Dataset and development environment ready
Phase 2	Month 2	Shared Vision Transformer encoder design and implementation, baseline reproduction of the separate-pathway hybrid Transformer-CNN framework	Encoder backbone implemented and baseline reproduced
Phase 3	Month 3	U-Net-style segmentation branch development, mask training and evaluation (Dice, IoU)	Working lesion segmentation model
Phase 4	Month 4	Classification branch training and evaluation (accuracy, AUC, sensitivity, specificity), Grad-CAM integration	Working classification model with explainability
Phase 5	Month 5	Clinical knowledge base construction, RAG-based retrieval, risk stratification and report-generation module	Clinical decision-support module completed
Phase 6	Month 6	Dashboard integration, end-to-end testing, performance evaluation and documentation	End-to-end prototype and final evaluation

6. EXPECTED OUTCOMES OF THE PROJECT:

The project is expected to produce a functional prototype for AI-driven skin lesion analysis, including a trained shared Vision Transformer-U-Net model for joint lesion segmentation and malignancy classification, together with Grad-CAM-based visual explanations for model transparency. The system will further provide a retrieval-augmented clinical decision-support module that links predictions to dermatological knowledge and guideline criteria, along with structured, referral-ready diagnostic reports. A comparative evaluation against the separate-pathway baseline framework will quantify gains in computational efficiency and predictive performance. The final outcome will be an integrated clinical decision-support dashboard for interpretable, AI-assisted skin cancer screening.

7. BUDGETARY REQUIREMENTS* (components & approximate costs in tabular form)

(*No funds for travel and equipment is allotted)

Sl. No.	Component	Approx. Cost
1	Cloud GPU / computational resources for Vision Transformer-U-Net training and experimentation	Rs. 8,000
2	Cloud storage and dataset backup for lesion images, masks and processed data	Rs. 2,000
3	Software, API and AI model services for explainability, RAG and clinical NLP experimentation	Rs. 3,000
4	Printing, documentation and prototype/presentation materials	Rs. 2,000
	TOTAL	Rs. 15,000

Existing institutional resources and open-source tools will be utilized wherever possible, while the proposed budget primarily supports computational resources, cloud storage, AI services, and project documentation.